Duothan 6.0

Beyond Coding, Into Engineering Excellence.

PHASE 02 · REBUILD

Build the platform . Prove it works.
Defend the system you designed, live in production.

duothan.ieeensbm.org

NSBM Green University IEEE Student Branch · IEEE Computer Society NSBM Green University Student Branch Chapter · IEEE SIGHT Special Interest Group on Humanitarian Technology NSBM Green University · NSBM Green University IEEE Student Branch

01 • CONTEXT

From Blueprint to Reality

A massive global cyber disaster in the year 2065, caused by the Super Malware Agent, has taken down systems across governments, businesses, healthcare, transportation, and financial services. The entire world has been brought to a standstill.

In the financial sector, the damage is severe. Core banking systems, ATMs, digital payment platforms, and loan systems have completely stopped working. Millions of people have lost access to essential financial services. Small businesses can no longer process digital transactions, and people are forced to rely entirely on cash. This loss of secure banking has slowed economic recovery and increased financial inequality across the world.

Fortunately, customer databases are safe because of secure backups, so no user information has been lost. However, the Master Key needed to unlock the banking network is hidden behind strong security layers. While other teams fight the malware in different sectors across the globe, the financial world needs a dedicated rescue effort.

The mission of Duothan 6.0 is to save the digital banking system

Designing, building, and deploying a secure, reliable, and inclusive financial platform from the ground up, using scalable architecture with **independent services**.

Rebuilding the old system is not enough. The new platform must use modern cybersecurity, cloud technology, and intelligent automation to make sure an attack like this can never happen again. It needs to restore daily banking operations, protect customer data, support secure transactions, and include a strong disaster recovery plan.

Your mission is to beat the malware, rebuild the financial ecosystem, and restore trusted digital banking for millions of users.

02 • BRIEF

Phase Overview

In Phase 2 - Rebuild, teams transition from planning to implementation by transforming their Phase 1 architecture and design document into a functional banking web application. This phase focuses on converting the proposed system architecture into a working solution that demonstrates the feasibility and practicality of the team's design. The application should reflect the objectives and features defined during the planning stage while maintaining consistency with the proposed architecture.

Teams are expected to develop the complete application by implementing the core banking functionalities, building both the frontend and backend components, and designing the required databases and APIs. The solution should provide seamless user experience, reliable system functionality, and effective communication between all application components. Throughout the development process, teams should ensure that their application is functional, organized, and aligned with the requirements established in Phase 1.

In addition, teams must manage their project using GitHub and follow proper software engineering practices, including version control, modular development and clean code organization. The final submission should demonstrate the team's ability to transform their architectural design into a complete, functional, and well-structured software solution that serves as the foundation for the final deployment phase.

03 · SUBMISSION

Final Deliverables

At the end of Phase 2, your team submits the following:

01	Public GitHub Repository Link
02	Working Web Application Source Code
03	User Guide Markdown File

SUBMISSION FORMAT

Teams must submit the **Public GitHub Repository Link** and the **Working Web Application Source Code Zip File** through the submission form. Ensure the GitHub repository is public and the source code is complete and functional before submission.

Submission Link : duothan.ieeensbm.org/submission

04 · EVALUATION

Mark Allocation

CRITERIA	WEIGHT
Solution’s functionality	15%
System Architecture & Best practices	15%
Client-Side handling	10%
Authentication system	15%
Server-side handling	20%
Quality Assurance Strategies	15%
Enterprise base Strategies	10%
TOTAL	100%

Note: the above weightings are indicative and may be adjusted by the judging panel to reflect the depth of execution shown across sections.

BY THE END OF PHASE 2

Your team should have transformed an idea into a working digital banking platform.

The blueprint created during RECON becomes reality during REBUILD.

The world needs a new financial ecosystem.

Build it. Secure it. Prepare it for the future.

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Questions? duothan.ieeensbm.org